

CrewAI with Real Web Search — Project Explanation

Overview

This project is the fifth entry in my Agentic AI series and a direct follow-up to an earlier CrewAI notebook. Where that notebook collected a Serper web search API key but never wired it into any agent, this project actually equips an agent with live web search, and introduces two genuinely different specialist agents collaborating in a sequential market-research pipeline rather than one agent handling every task.

What the Notebook Does

A Market Researcher agent is given a real SerperDevTool for live internet search, while a Product Strategist agent is deliberately given no tools at all, modelling real division of labour where not every agent in a crew needs identical capabilities. Two tasks are defined: the researcher's task browses the internet for current market trends around a product launch, and the strategist's task builds a positioning strategy explicitly described as being based on those market insights.

Under CrewAI's default sequential process, the second task automatically receives the first task's output as context, with no explicit conversation-history management required, a genuinely different mental model from the AutoGen-based notebooks earlier in this series where passing information between agents means managing message history directly. The crew is assembled with planning enabled and run asynchronously. The accompanying documentation also flags a real bug found in the crew-assembly code, a capitalised Agents keyword argument where CrewAI expects a lowercase agents, which would raise an error as written.

Skills Demonstrated

- Equipping individual agents with distinct tools rather than a shared generic toolset
- Understanding CrewAI's automatic sequential context-passing between tasks
- Integrating live web search into an agentic pipeline via crewai-tools
- Debugging keyword-argument errors in multi-agent orchestration code

This tool-equipped, sequentially-dependent agent pipeline is a direct step toward the specialised, information-gathering agent roles my dissertation's agentic intelligence platform relies on.